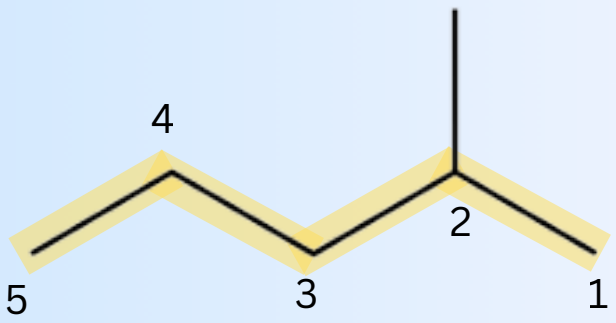
CARD A01 — STRUCTURE → NAME

Level: Foundation Molecular Formula: C_6H_{14}

1. Numbered Parent



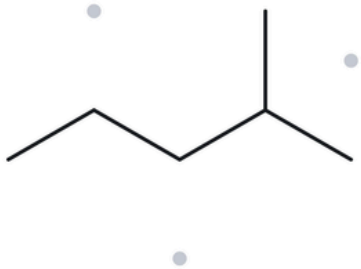
2. Naming Trail

Parent: longest chain → 5C, pentane. Other paths exist but gives the same result for methyl placement.
Numbering: L→R gives 4, R→L gives 2 for methyl location. R→L it is.
Substituents: methyl on C2→ 2-methyl

3. My Name

2-methylpentane

4. The Alkane Check



2-methylpentane

Returned name:
2-methylpentane

✓ Match

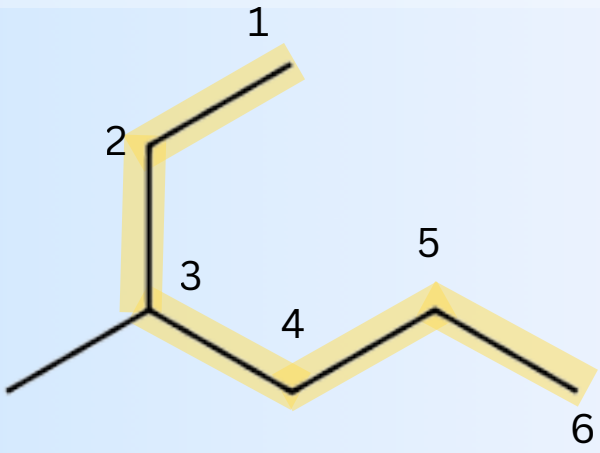
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A02 — STRUCTURE → NAME

Level: Foundation Molecular Formula: C_7H_{16}

1. Numbered Parent



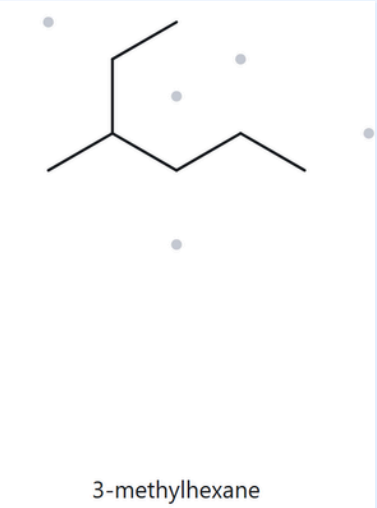
2. Naming Trail

Parent: longest chain → 6C, hexane. The only chain giving the longest.
Numbering: U→R gives 3, R→U gives 4 for methyl location. U→R it is.
Substituents: methyl on C3→ 3-methyl

3. My Name

3-methylhexane

4. The Alkane Check



Returned name:
3-methylhexane

✓ Match

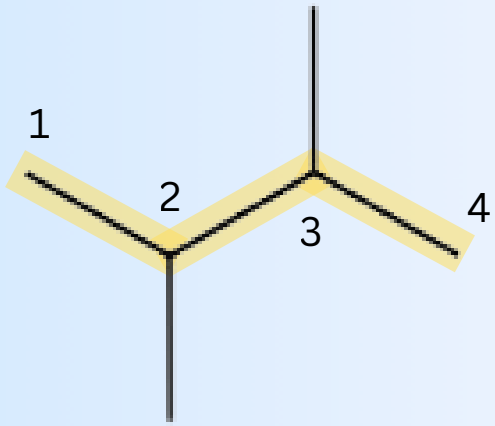
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A03 — STRUCTURE → NAME

Level: Foundation Molecular Formula: C_6H_{14}

1. Numbered Parent



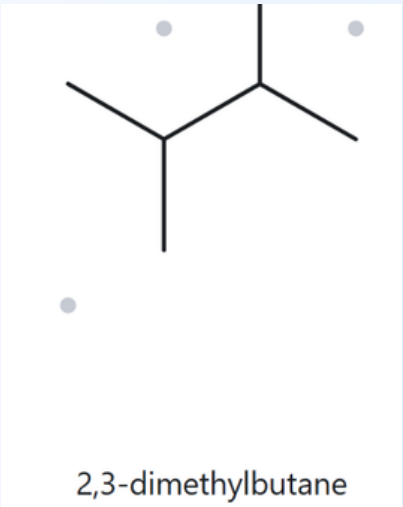
2. Naming Trail

Parent: longest chain → 4C, butane. Other paths exist but gives the same result for methyl locants.
Numbering: L→R gives 2,3, R→L gives 2,3 also.
Substituents: methyl on C2, C3→ 2,3-dimethyl

3. My Name

2,3-dimethylbutane

4. The Alkane Check



Returned name:
2,3-dimethylbutane

✓ Match

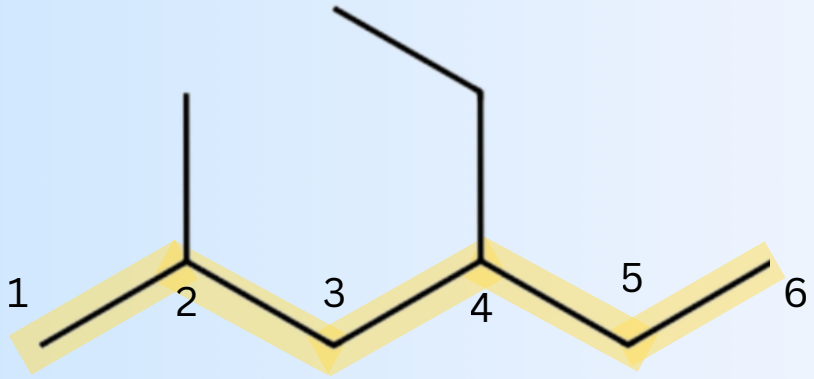
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A04 — STRUCTURE → NAME

Level: Developing Molecular Formula: C_9H_{20}

1. Numbered Parent



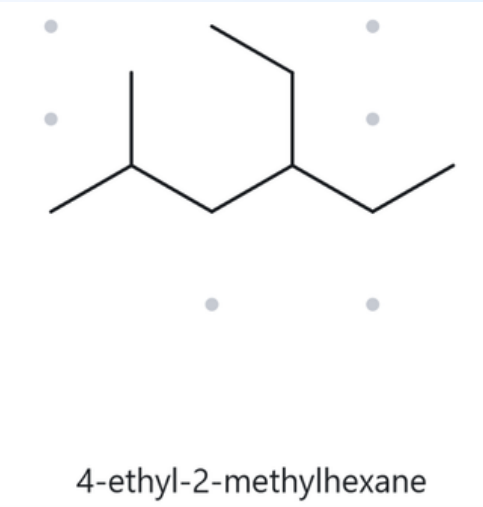
2. Naming Trail

Parent: longest chain → 6C, hexane. Other paths exist but have farther locant for first substituent.
Numbering: L→R gives 2,4, R→L gives 3,5 for methyl location. L→R it is.
Substituents: ethyl on C4, methyl on C2→
Alphabetization: 4-ethyl-2-methyl

3. My Name

4-ethyl-2-methylhexane

4. The Alkane Check



Returned name:
4-ethyl-2-methylhexane

✓ Match

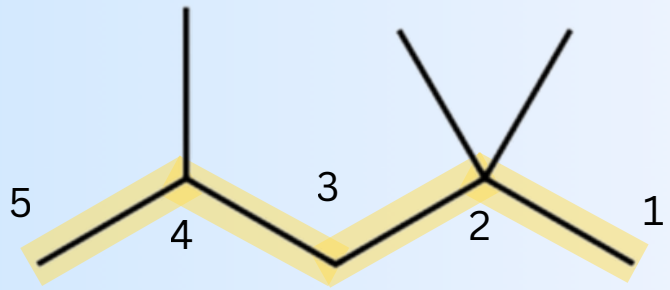
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A05 — STRUCTURE → NAME

Level: Developing Molecular Formula: C_8H_{18}

1. Numbered Parent



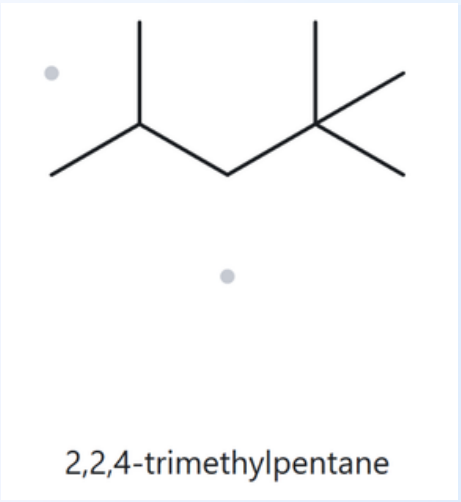
2. Naming Trail

Parent: longest chain → 5C, pentane. Other paths exist but have farther locant for first substituent.
Numbering: L→R gives 2,4,4, R→L gives 2,2,4. By first point of difference, it is R→L.
Substituents: dimethyl on C2, methyl on C4→ 2,2,4-trimethyl

3. My Name

2,2,4-trimethylpentane

4. The Alkane Check



Returned name:
2,2,4-trimethylpentane

✓ Match

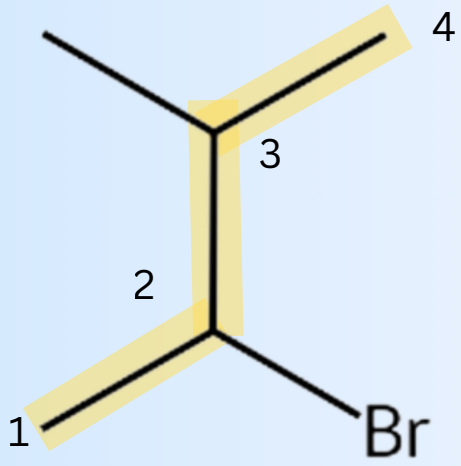
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A06 — STRUCTURE → NAME

Level: Developing Molecular Formula: $C_5H_{10}Br$

1. Numbered Parent



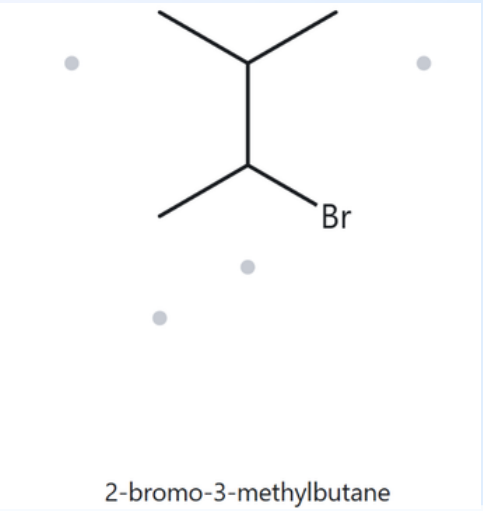
2. Naming Trail

Parent: longest chain → 4C, butane. Other paths exist but have farther locant for first(bromo) substituent.
Numbering: L→R gives 2,3, R→L gives 2,3 also but with **b**romine first.
Substituents: bromine on C2, methyl on C3→ 2-bromo-3-methyl

3. My Name

2-bromo-3-methylbutane

4. The Alkane Check



Returned name:
2-bromo-3-methylbutane

✓ Match

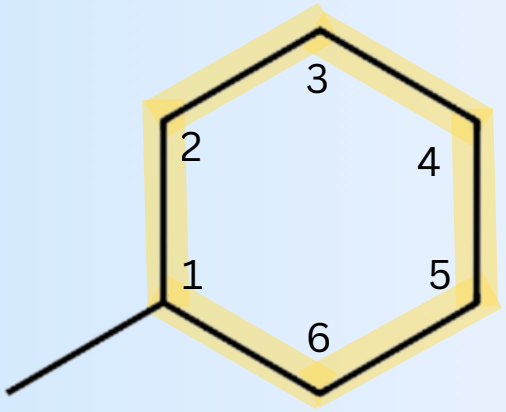
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A07 — STRUCTURE → NAME

Level: Developing Molecular Formula: C_7H_{14}

1. Numbered Parent



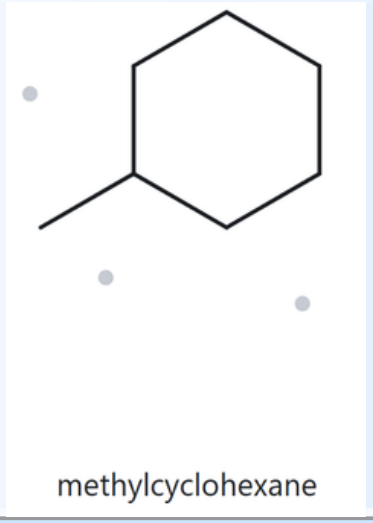
2. Naming Trail

Parent: longest chain → 6C ring, cyclohexane. Ring exists and there is no chain.
Numbering: Where the substituent starts.
Substituents: methyl on C1→ methyl

3. My Name

methylcyclohexane

4. The Alkane Check



Returned name:
methylcyclohexane

✓ Match

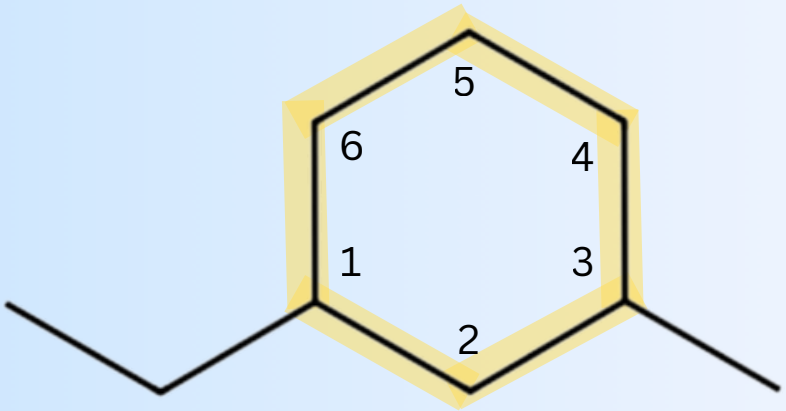
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A08 — STRUCTURE → NAME

Level: Challenge Molecular Formula: C_9H_{18}

1. Numbered Parent



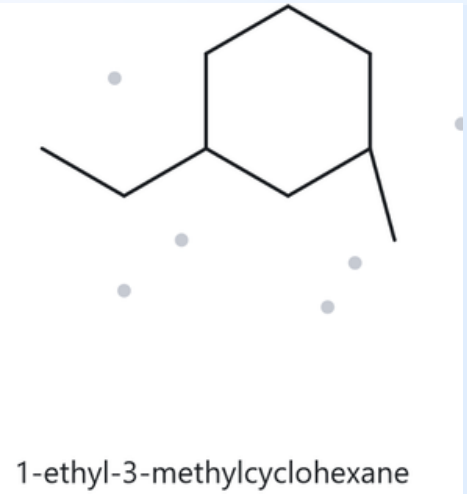
2. Naming Trail

Parent: longest chain → 6C ring, cyclohexane. The ring has more carbon than the chain.
Numbering: From where **e**thyl starts, to nearest methy locant.
Substituents: ethyl on C1, methyl on C3→ 1-ethyl-3-methyl

3. My Name

1-ethyl-3-methylcyclohexane

4. The Alkane Check



Returned name:
1-ethyl-3-methylcyclohexane

✓ Match

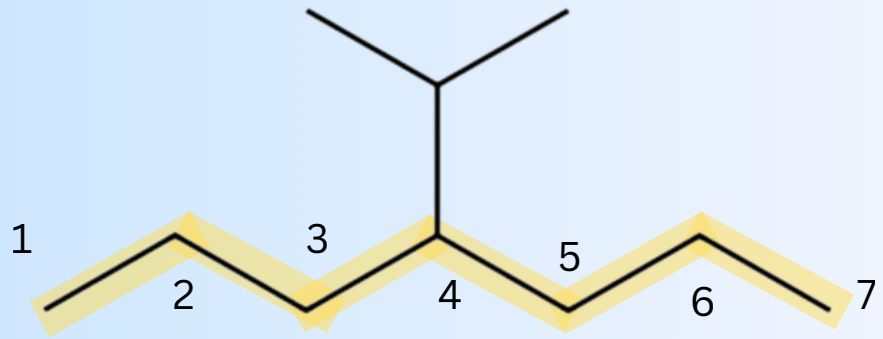
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A09 — STRUCTURE → NAME

Level: Challenge Molecular Formula: $C_{10}H_{22}$

1. Numbered Parent



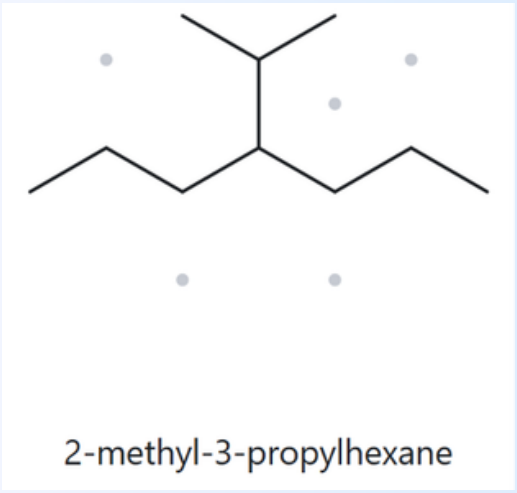
2. Naming Trail

Parent: longest chain → 7C, heptane. Other paths exist but gives the same result for methyl placement.
Numbering: Both L→R and R→L gives 4, so either one.
Substituents: isopropyl on C4→ 4-isopropyl

3. My Name

4-isopropylheptane

4. The Alkane Check



Returned name:
2-methyl-3-propylhexane

❌ Unmatched

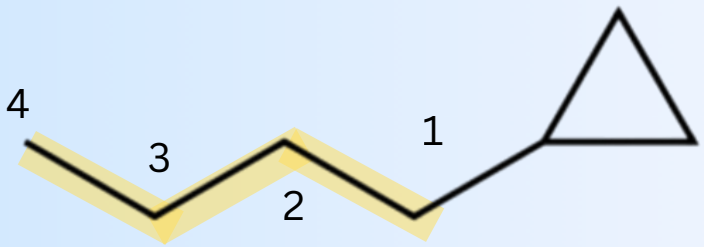
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A10 — STRUCTURE → NAME

Level: Challenge Molecular Formula: C_7H_{14}

1. Numbered Parent



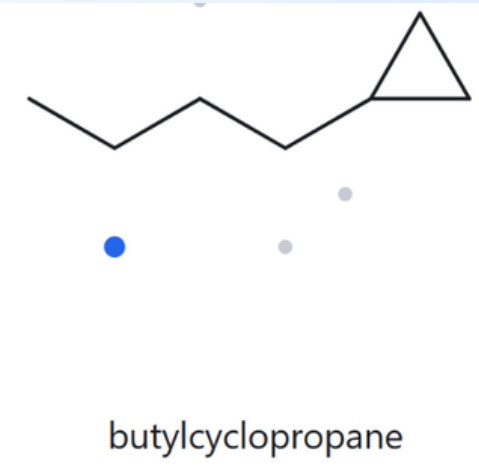
2. Naming Trail

Parent: longest chain → 4C, butane. A ring exists but has less carbon than the chain.
Numbering: L→R gives 4, R→L gives 1 for cyclopropyl location. R→L it is.
Substituents: cyclopropyl on C1→ cycropropyl

3. My Name

cyclopropylbutane

4. The Alkane Check



Returned name:
butylcyclopropane

❌ Unmatched

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

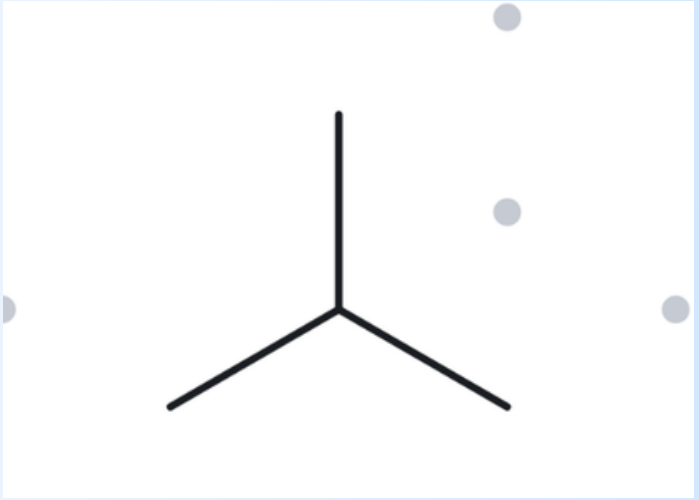
CARD B01 — NAME→ STRUCTURE

Level: Foundation Molecular Formula: C_4H_{10}

1. Name Decoded

Parent: propane → 3C chain
Substituents: methyl on C2
Expected Formula: 3+1= 4C → C_nH_{2n+2} = C_4H_{10}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=4, H=10 → C_4H_{10}

4. Back Check

Returned by The Alkane:
2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

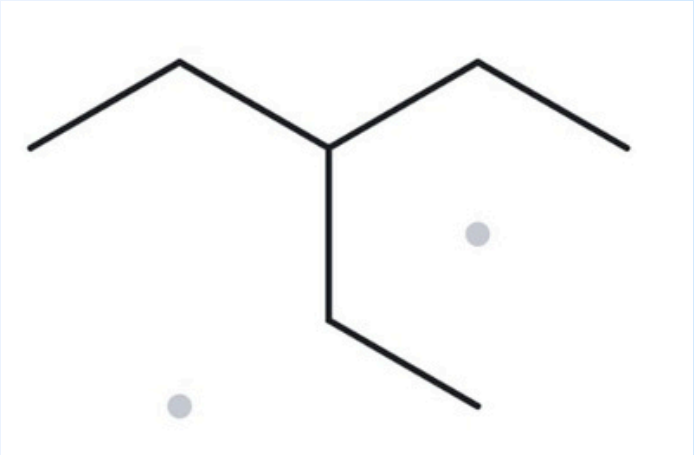
CARD B02 — NAME→ STRUCTURE

Level: Foundation Molecular Formula: C_7H_{16}

1. Name Decoded

Parent: pentane → 5C chain
Substituents: ethyl on C3
Expected Formula: 5+2= 7C → C_nH_{2n+2} = C_7H_{16}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=7, H=16 → C_7H_{16}

4. Back Check

Returned by The Alkane:
3-ethylpentane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

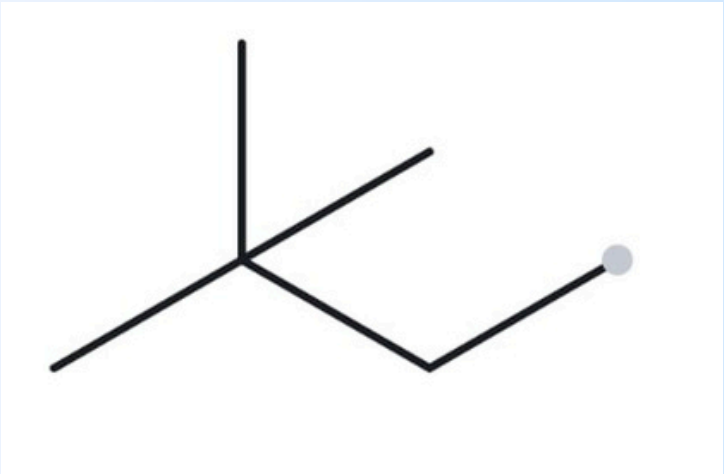
CARD B03 — NAME→ STRUCTURE

Level: Foundation Molecular Formula: C_6H_{15}

1. Name Decoded

Parent: butane → 4C chain
Substituents: dimethyl on C2
Expected Formula: 4+2= 6C → C_nH_{2n+2+1} = C_6H_{15}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=6, H=15 → C_6H_{15}

4. Back Check

Returned by The Alkane:
2,2-dimethylbutane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

CARD B04 — NAME→ STRUCTURE

Level: Developing Molecular Formula: C_8H_{18}

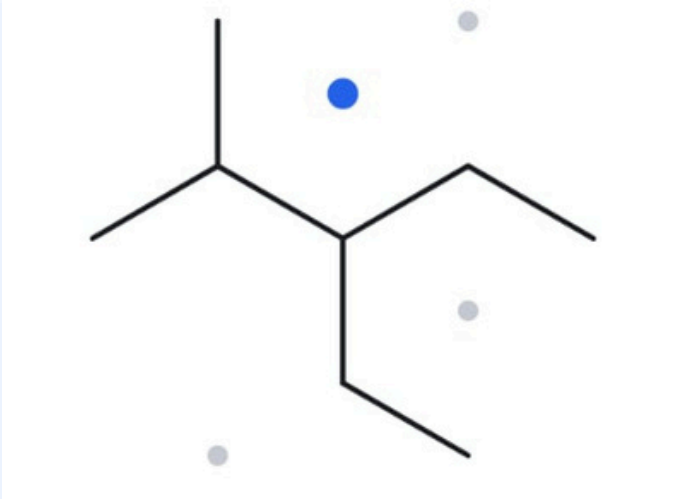
1. Name Decoded

Parent: pentane → 5C chain

Substituents: methyl on C2, ethyl on C3

Expected Formula: 5+1+2= 8C → C_nH_{2n+2} = C_8H_{18} .

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=8, H=18 → C_8H_{18}

4. Back Check

Returned by The Alkane:

3-ethyl-2-methylpentane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

CARD B05 — NAME→ STRUCTURE

Level: Developing Molecular Formula: C_4H_9Cl

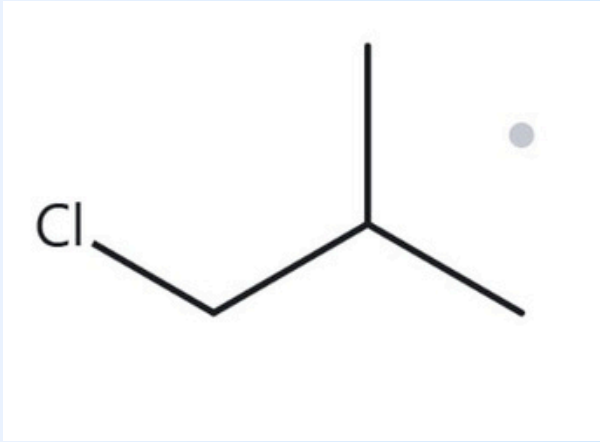
1. Name Decoded

Parent: propane → 3C chain

Substituents: chloro on C1, methyl on C2

Expected Formula: 3+1= 4C → C_nH_{2n+2-3} Cl= C_4H_9Cl

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=4, H=9 , Cl=1→ C_4H_9Cl

4. Back Check

Returned by The Alkane:

1-chloro-2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

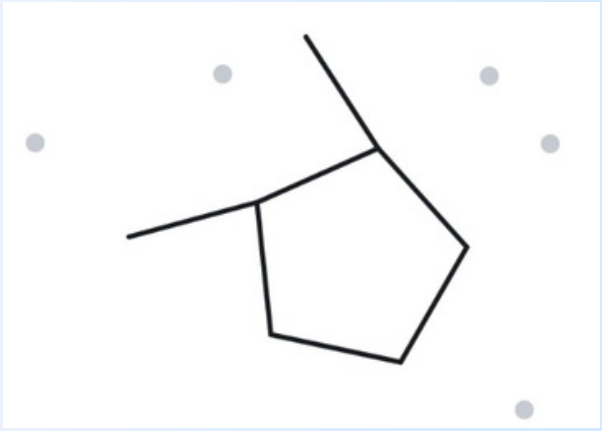
CARD B06 — NAME→ STRUCTURE

Level: Developing Molecular Formula: C_7H_{14}

1. Name Decoded

Parent: cyclopentane → 5C ring
Substituents: methyl on C1 and C2
Expected Formula: 5+2= 7C → C_nH_{2n}= C_7H_{14}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=7, H=14 → C_7H_{14}

4. Back Check

Returned by The Alkane:
1,2-dimethylcyclopentane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

CARD B07 — NAME→ STRUCTURE

Level: Developing Molecular Formula: $C_{11}H_{24}$

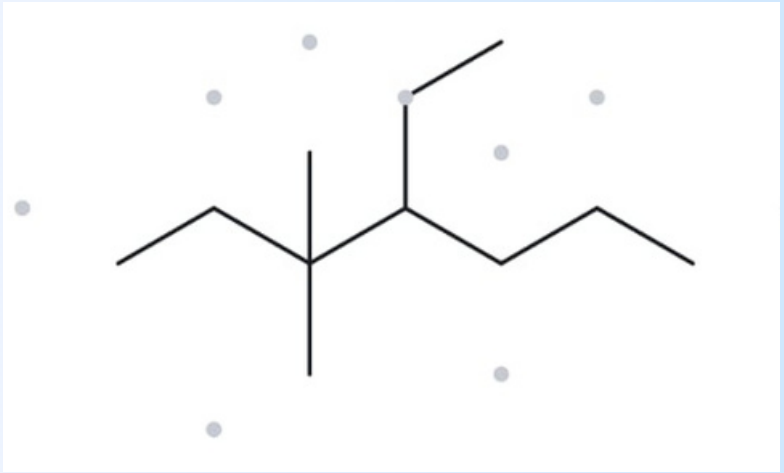
1. Name Decoded

Parent: heptane → 7C chain

Substituents: dimethyl on C3, ethyl on C4

Expected Formula: 7+2+2= 11C → C_nH_{2n+2}= $C_{11}H_{24}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=11, H=24 → $C_{11}H_{24}$

4. Back Check

Returned by The Alkane:

4-ethyl-3,3-dimethylheptane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

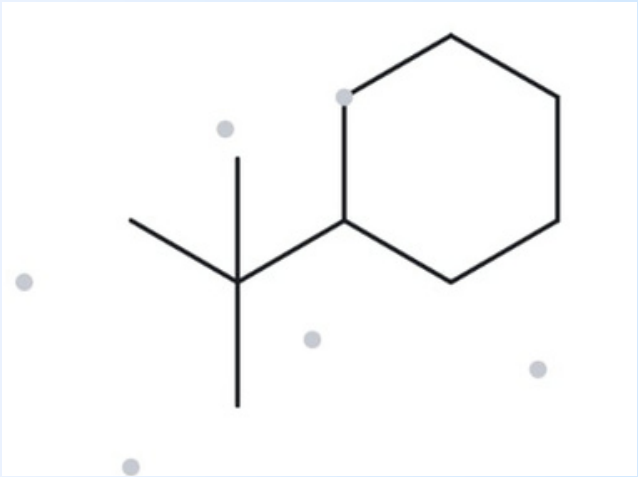
CARD B08 — NAME→ STRUCTURE

Level: Challenge Molecular Formula: $C_{10}H_{20}$

1. Name Decoded

Parent: cyclohexane → 6C ring
Substituents: ethyl and dimethyl on C1
Expected Formula: 6+2+2= 10C → C_nH_{2n}= $C_{10}H_{20}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=10, H=20 → $C_{10}H_{20}$

4. Back Check

Returned by The Alkane:
(1,1-dimethylethyl)cyclohexane
is equal to tert-butylcylohexane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

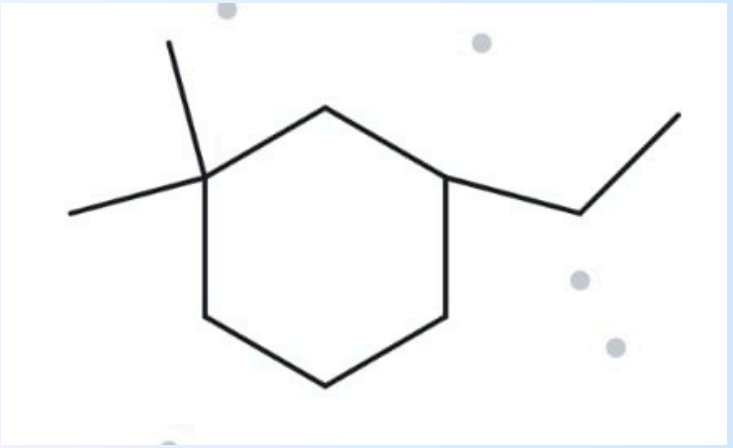
CARD B09 — NAME→ STRUCTURE

Level: Challenge Molecular Formula: $C_{10}H_{20}$

1. Name Decoded

Parent: cyclohexane → 6C ring
Substituents: dimethyl on C1, ethyl on C3
Expected Formula: 6+2+2= 10C → C_nH_{2n}= $C_{10}H_{20}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=10, H=20 → $C_{10}H_{20}$

4. Back Check

Returned by The Alkane:
3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded

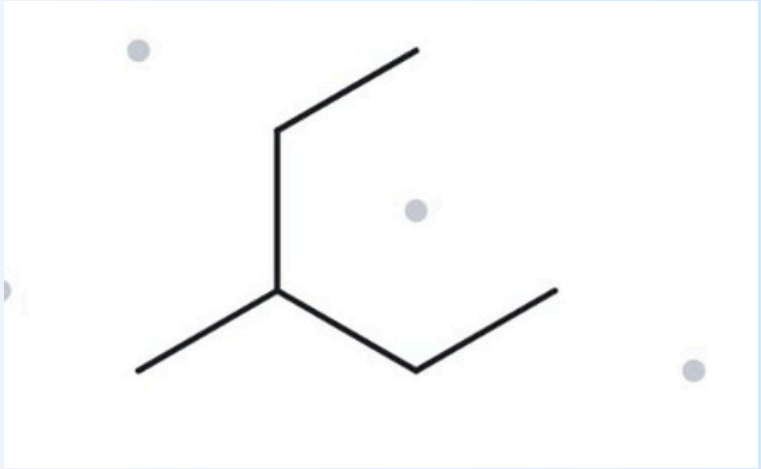
CARD B010 — NAME→ STRUCTURE

Level: Challenge Molecular Formula: C_6H_{14}

1. Name Decoded

Given: 2-ethylbutane
Correct name: 3-methylpentane
Parent: pentane → 5C chain
Substituents: methyl on C3
Expected Formula: 5+1= 6C → C_nH_{2n+2}= C_6H_{14}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the Drawing: C=6, H=14 → C_6H_{14}

4. Back Check

Returned by The Alkane:
3-methylpentane

✓ Same as the correct name

VERIFICATION STRIP

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-check recorded